

Research Module in Applied Geophysics:
Report

Research Project WS25/26

Investigating Cable-Ground Coupling of Unburied DAS for Lunar
Applications

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Abstract

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1 Introduction

The robustness and long aperture capabilities of Distributed Acoustic Sensing (DAS) make it a promising seismic sensor for detecting moonquakes in the harsh lunar environment (Wu et al., 2024). Collecting more seismological data on our celestial companion would enable us to refine our models of its planetary structure and history. It might be impractical to bury fiber optic cables for long arrays, so gaining more insight into the transfer function of DAS on loosely coupled materials could enable more accurate quantification of seismic signals on the Moon. The objective of the upcoming master's thesis is to understand the physical mechanisms of loosely coupled fiber optic cables used in DAS in relation to ground deformation. This will be accomplished by deriving transfer functions for the DAS system and observing how they change with different surface materials, coupling, or fiber types. The thesis will also evaluate DAS based on different wave field characteristics.

The transfer function is determined by comparing the DAS strain rate response to the actual ground motion. Experimental measurements for quantification will be conducted at the ETH Zurich Center for Immersive Wave Experimentation (CIWE). CIWE's laboratory incorporates a robotized 3D scanning laser Doppler vibrometer (LDV), which can precisely measure the velocity of a point in three directions. A designated source will be used to excite a seismic wave field. A designated source will be used to generate a seismic wave field. Since the LDV can only measure at one location at a time and DAS works like an array along the fiber, the source must be activated repeatedly so the LDV can scan multiple points along the line. Figure 1 illustrates what an experiment like this would look like. To prepare for this thesis, we conducted literature research on DAS and programmed an initial processing scheme for LDV data. The first part of this report will present the concepts of DAS, describe relevant parameters, and analyze performance. The second part of the study is an analysis of an LDV dataset from 2024. Standard signal processing workflows were used to analyze the data.

2 Review: Distributed Acoustic Sensing

2.1 Concept

Types of DAS. Give examples. We will focus on Optical-Time-Domain Reflectometry.

2.1.1 Gauge Length

2.1.2 Pulse Width and Shape

Pulse shape can depend on the optoelectronics (Hubbard 2022) but are approximately a gaussian (Dean et al. 2016).

2.1.3 Directionality

2.2 Performance

2.2.1 Instrument Response

A description of the transfer function of DAS with respect to gauge length and pulse shape was given by Hubbard (2022). In this case a perfectly coupled fiber was assumed. As the total transfer function is a multiplication of two transfer functions the result depends on individual whatever.

The theoretical amplitude and phase response of DAS is flat. Lindsey et al. (2020) proved ex-

perimentally using teleseismic and ocean micro-seismic data that this holds in the range of 10 - 120 seconds. For periods shorter than that (1 - 10 seconds) the amplitude response of DAS is 3–11 dB higher above the actual ground movement. BUT this might be due to coupling conditions of the experiment (confirm)

2.2.2 Cable Coupling

2.2.3 Limitations

Several factors play into the role of the instrument response for DAS.

EAGE Dean states, that the gauge length has a smoothing effect on the wavelet. But its not just the gauge length. During acquisition a gaussian window is applied at each end of the gauge length, resulting in further smoothing.

There is a strong relation between the correct gauge length and pulse width to the SNR. For example, synthetic data shows that the SNR is quite low when the pulse width is much smaller than the gauge length. If the pulse width is e.g. half of the gauge length then it is more stable (Dean et al...). At this point the effect of the pulse width on the frequency response becomes greater (i think).

The gauge length has a strong influence on the frequency response.

3 LDV Dataset

3.1 Overview

In the following section, we will investigate an experiment in which a laser Doppler vibrometer (LDV) was used to collect seismic data. This data comes from an experiment ... (Q: ask Simone for details). As illustrated in Figure 1, the LDV scanned along a line on an 2 m x 30 mm x 30 mm aluminum beam. It recorded the particle velocity at 100 measurement points spaced 20 mm [dx] apart. The three-dimensional (3D) measurement directions are depicted by the LDV coordinate system on the left-hand side of the beam. The x-component is parallel to the beam, the y-component is perpendicular to the beam's length, and the z-component points upward. A piezoelectric transducer was attached to one end of the beam. Ten milliseconds after the start of the recording, it excited a Ricker wavelet signal with a center frequency of 5 kHz in the x-direction. The LDV sampled data at 200 kHz. A 400 ms recording time resulted in 80,000 samples being recorded at each location in each direction. The source was activated twice per second. Data was recorded nine times for each location, summed up, and averaged for the final dataset. All acquisition parameters are summarized in Table 1.

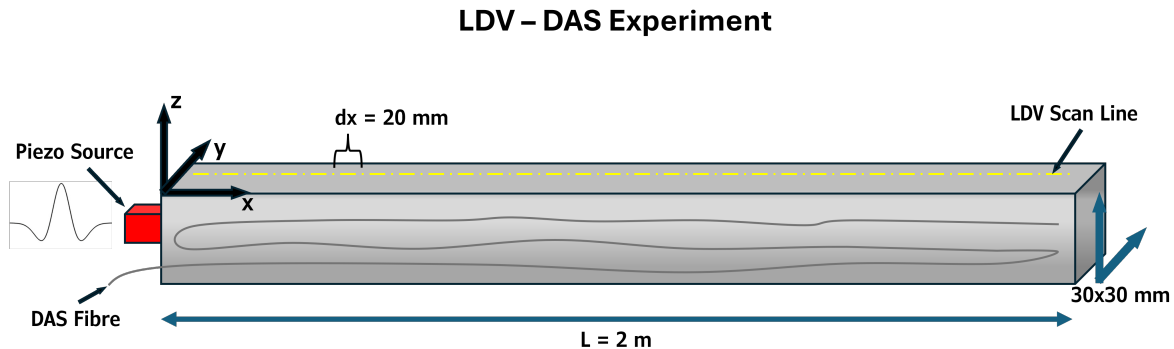


Figure 1: Sketch of the experiment setup

Parameter	Value
Profile Length [L]	2 m
Sample Spacing [dx]	0.02 m
Number of Sampling Points	100
Sampling Rate	200 kHz
Recording Length [T]	400 ms
Source Wavelet, Center Frequency	Ricker, 5 kHz
Trigger Time	10 ms
Number of Stacks	9

Table 1: LDV Measurement Parameters

3.2 Processing

We processed the LDV dataset using the programming language MATLAB. The processing scripts can be found under [Cornelius \(2026\)](#). (Q: Is it allowed to put the code on a public repository, since that would be pre-publication?). To visualize wave field propagation, an animation was created depicting material velocity in space, with time as the animating variable. Figure 2 depicts the vibration amplitude along the beam 10.30 ms after the start of the measurement. The red graph at the top, the green graph in the middle, and the blue graph at the bottom depict the x-, y-, and z-components, respectively.

A metal beam acts as a waveguide for seismic waves. Under these conditions, two types of waves are primarily observed: Longitudinal waves, in which particles oscillate parallel to the direction of propagation, causing compression, and flexural waves, which involve transverse displacement perpendicular to the direction of propagation. Flexural waves are often dispersive, meaning higher frequencies travel faster ([Ernst et al., 2016](#)).

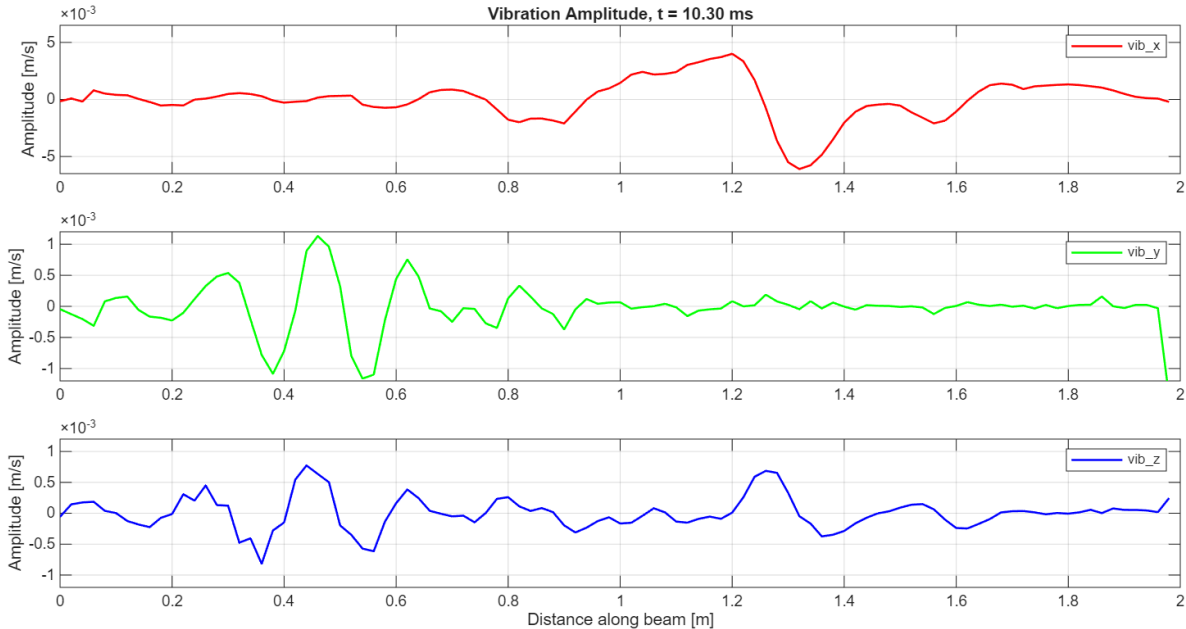


Figure 2: Vibration amplitudes along the beam 0.30 ms after source excitation

In the x-direction, we observe that the wavelet arrives with a small negative amplitude pulse that returns to zero ground velocity. Then, it drops to larger negative amplitudes of around -5 mm/s before turning positive and reaching nearly 5 mm/s. Eventually, it flattens out. By that time, the wave traveled around 1.5 m. The y-component shows the flexural wave, which resembles a Morlet wavelet.

It has a positive peak in the center and side lobes with lower amplitudes on each side. At the time of its first arrival, it had traveled approximately 0.8 m. Both wave types are visible on the z-component, but the longitudinal wave is slightly ahead of the x-component. The maximum amplitude of the y-component is approximately five times lower than that of the x-component. The z-component amplitude is even smaller than that.

Another way to visualize seismic data is with a shot gather, which plots each location as a seismic trace over time. Figure 3 depicts the recorded traces of measurements taken between 9.8 and 11 ms. Figure 3a shows the data plotted as wiggles, while Figure 3b shows them as an image, where amplitude is displayed using a grayscale. Each trace is normalized. As we can see, the longitudinal wave propagates faster and reflects at around 10.4 ms from the end of the beam. The flexural wave reflection is outside the time window, but can be seen in Figure 4. Some noisier traces can be identified in the shot gather. Examples can be seen at the beginning and end of the beam. (Q: Should I mark them in the Figure?). The shot gather plots for the other components can be found in the appendix.

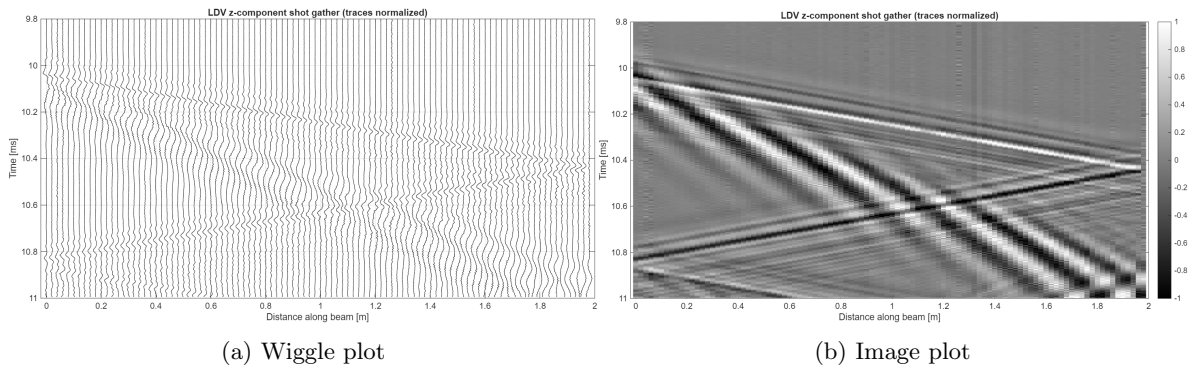


Figure 3: Shot gather of the z-component depicted as a wiggle and image plot from 9.8 to 11 ms.

From the shot gather plots, we can see that the phase of the longitudinal wave shifts slowly during propagation. Meanwhile, the phase velocity of the flexural wave noticeably exceeds the group velocity. This results in a striped pattern. It is also evident that the flexural wave widens over time during propagation. Figure 4 depicts the shot gather of the y-component from 9.8 to 11.8 ms. In this plot, we can see how the high frequencies are moving at a higher velocity than the low frequencies. Around 10.8 ms, when the longitudinal wave reflects for the second time at the beginning of the beam, another flexural wave is excited. The reason this excitation did not occur at the first reflection, but only at the second, may be due to interactions with the piezo element connected to the beam.

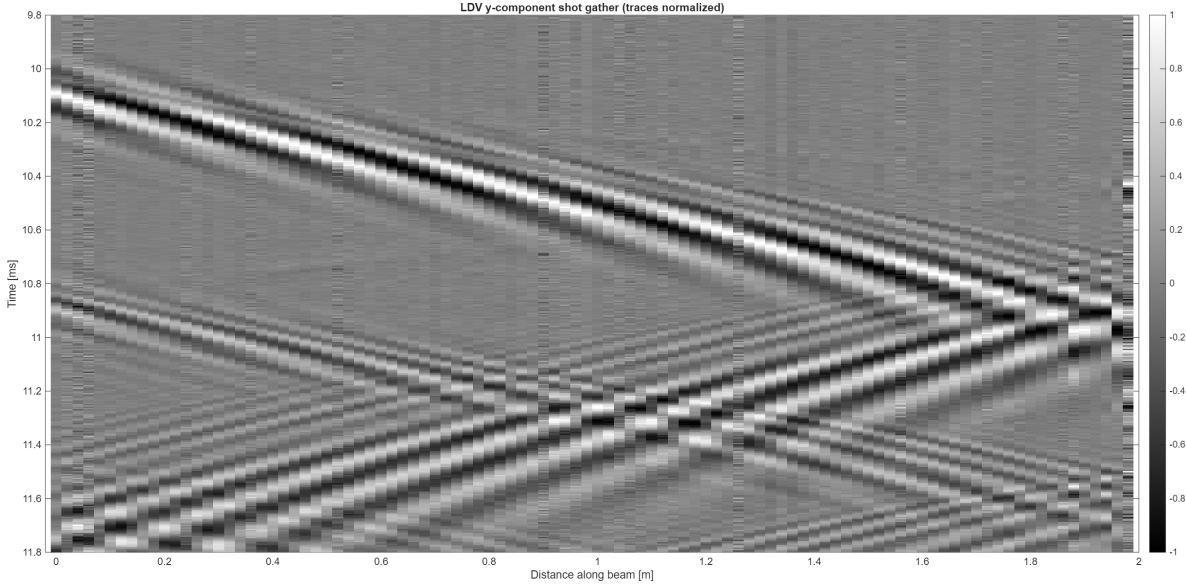


Figure 4: Shot gather of the z-component depicted as a wiggle and image plot from 9.8 to 11.8 ms.

3.2.1 Amplitude and Phase Spectra

For the amplitude and phase spectrum analysis, we included a 3 ms time window before triggering the source. This increased the spectral resolution without allowing the frequency content of noisy traces to dominate. We set the end of the window at 10.6 ms. A visual inspection of the shot gather revealed that, during that time, both dominant waves were fully within the frame. Since the longitudinal wave had already been reflected by that time, we attempted to shorten the window to 10.4 ms. However, since there was no significant difference in the spectrum, we kept it at 10.6 ms for consistency. We investigated the spectral content of the data by performing a Fourier transform on each trace. Then, we added up the amplitude spectra and calculated the mean. Figure 5 shows the normalized spectra of all components. The amplitude spectrum of the x-component (see Fig. 5a) consists of one primary peak between 3 and 17 kHz. The maximum is at 7.8 kHz. There is a plateau from 11 to 13 kHz. From 13 to 24 kHz, a second, smaller peak appears, with its maximum around 22 kHz. The y-component spectrum has similar characteristics (see Fig. 5b). The main peak is present with a maximum at 7.2 kHz, but the second, higher-frequency peak is significantly lower. This second peak is more prominent for the z-component (see Fig. 5c). The maximum of its main peak is at 6.4 kHz.

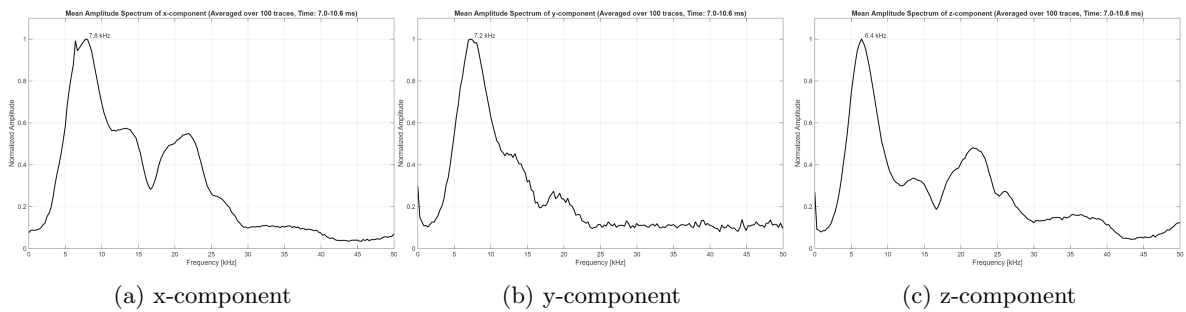


Figure 5: Mean amplitude spectra of all traces from 7 to 10.6 ms. The maximum is marked.

The phase spectrum analysis was performed on a single trace. We analyzed the trace recorded at 0.5 m. At that position, the signal-to-noise ratio (SNR) is high, and by $t = 10.6$ ms, both types of waves had already passed that position. Figure 6 displays the three components amplitude spectra in blue and the phase spectra in red. The amplitude spectra appear similar to those in Figure 5, but

are less smooth. Note that phase information for frequencies with low amplitudes is unreliable. The phase spectra have a sawtooth or ramp waveform that fluctuates between $-\pi$ and π . There is no frequency bin that lands exactly on $\pm\pi$, which may be due to the discrete sampling. The ramps of the x-component are closer together than those of the y-component. Since both the longitudinal and flexural waves are visible on the z-component, the phase spectrum depicts a mix of wider ramps for low frequencies and tighter ones for higher frequencies. (Q: I don't really know how to interpret these phase spectra. I also tried to plot them just for the wave-pulse (no time before and after) but that didn't give me more insight...)

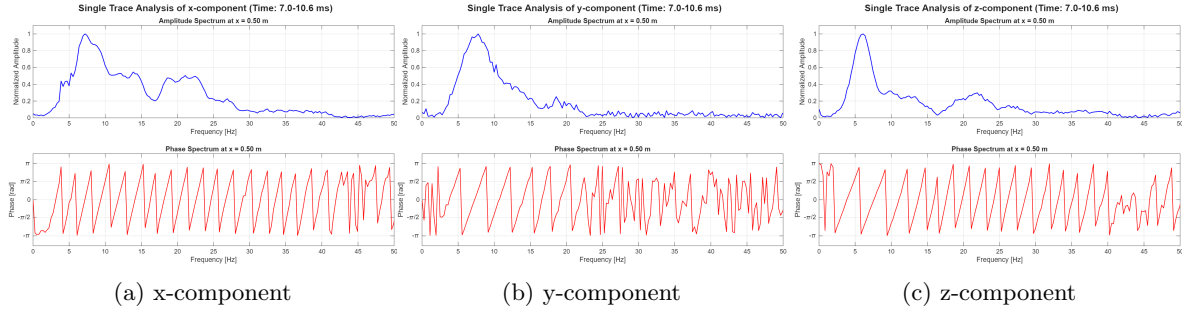


Figure 6: Amplitude and phase spectra at 0.5 m from 7 to 10.6 ms

3.2.2 Spectrograms

Subsequently, we computed spectrograms of the traces. Figure 7 depicts the spectrogram of the z-component at 0.5 m. Observe that the second frequency peak, around 22 kHz, arrives earlier than the primary peak, around 7 kHz. This corresponds to the fact that the second peak only contains the faster longitudinal wave, whereas the primary peak contains both waves. Higher modes are visible between 45 and 60 kHz and above 80 kHz. Spectrograms of the other components can be found in the appendix.

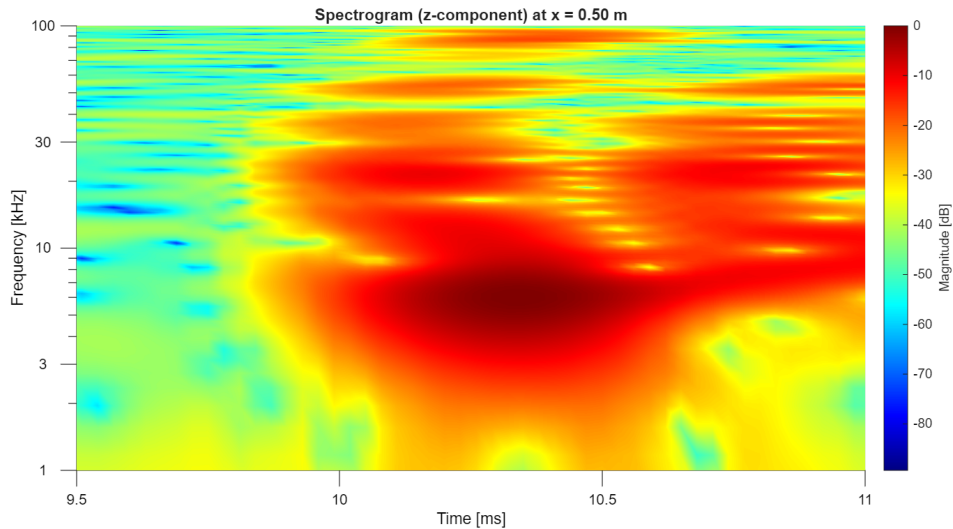


Figure 7: Spectrogram of the z-component at 0.5 m from 9.5 to 11 ms

3.2.3 f-k Spectra

By performing a double Fourier transform over time and space, we calculated frequency-wavenumber (f-k) spectra. Frequencies correspond to oscillation in time, while wavenumber describes oscillation in

space. Since the signals are real-valued, the amplitude spectra are symmetrical, and we can discard negative frequencies. However, due to directional propagation, we must consider both positive and negative wavenumbers. Figure 8 shows the shot gather and f-k spectrum of the longitudinal wave propagating in the positive direction. Figure 9 shows the same plots for the wave after it is reflected at the beam boundary and changes direction. In the first case, the wavenumbers are positive and range from 0 to approximately 70 cycles per meter. In the f-k domain, it appears as a linear feature. For the backward-propagating wave, the feature looks similar, but it is mirrored on the frequency axis. Hence, it consists primarily of negative wavenumbers. Since the time window is shorter than in the first case, the frequency resolution is slightly lower. The appendix contains all components of the f-k spectra for a time window of 9.8 to 11 ms.

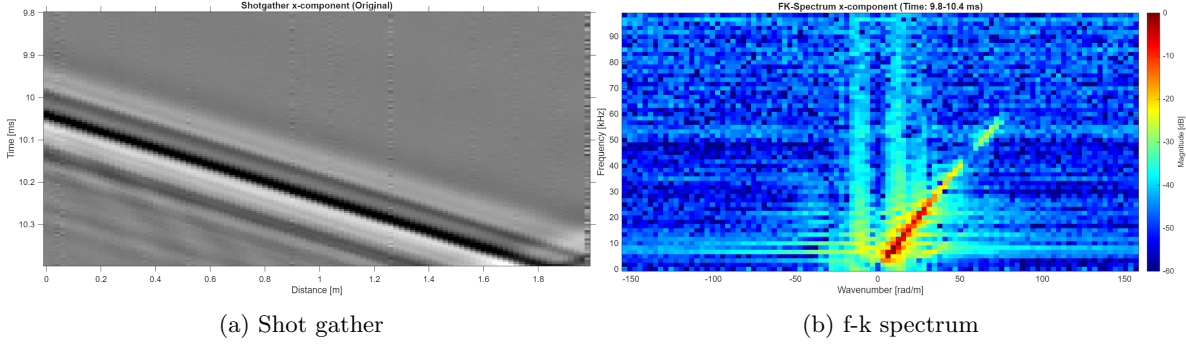


Figure 8: Shot gather and f-k spectrum of the forward propagating longitudinal wave (9.8 to 10.4 ms)

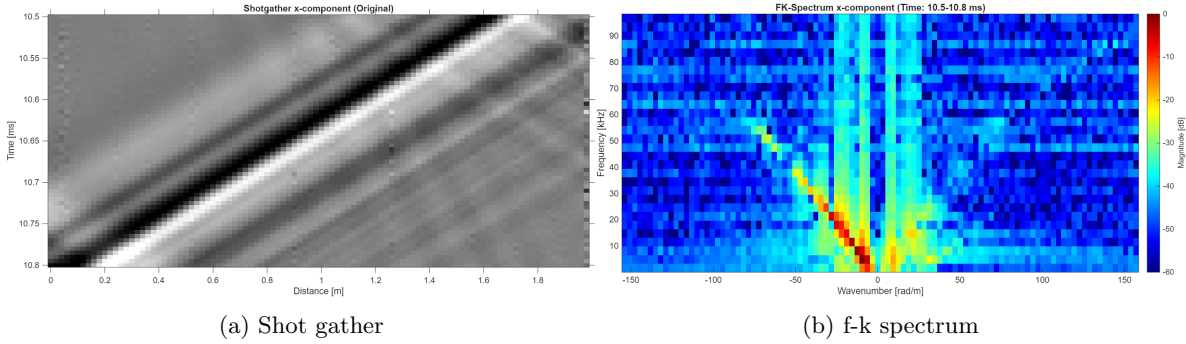


Figure 9: Shot gather and f-k spectrum of the reflected, backward propagating longitudinal wave (9.8 to 10.4 ms)

On the left side of Figures 10b and 11b, the f-k spectrum for the z-component is plotted. The linear feature of the longitudinal wave that we observed in the x-component appears even longer here and extends to higher wavenumbers and frequencies. At these higher frequencies, the linear feature appears to curve slightly downward towards the wavenumber axis. At around 90 Hz, it reaches the boundary and continues on the top left side of the plot. This artifact is due to spatial aliasing. Additionally, we can see a second linear feature that is less steep in the f-k domain and consists of a lower frequency and wavenumber content. In the plot, it mainly exists on the positive wavenumber side, which is due to the fact that the reflection of flexural waves is not entirely captured in the chosen time window.

The relationship between velocity c , frequency f , and wavenumber k is given by $c = f/k$. Using this, we can apply a filtering mask to the f-k spectra and filter out content that is higher or lower than a certain velocity. Figures 10 and 11 depict, on the left, the shot gathers and, on the right, the f-k spectrum before and after applying high- and low-pass velocity filters of 3000 m/s, respectively. To prevent ringing, a 20% cosine taper was implemented to the filter mask. After filtering out velocities under 3000 m/s, the image plot shows that only the longitudinal wave remains (see Fig. 10a on the right). In the opposite case, only the flexural wave remains (see Fig. 11a on the right).

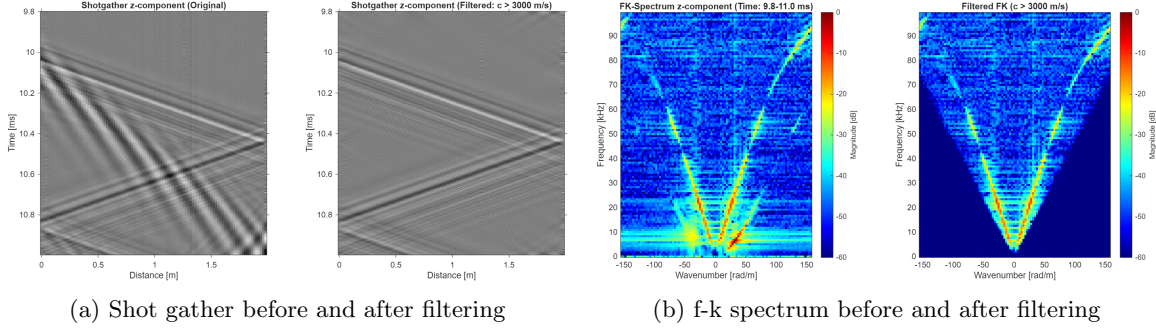


Figure 10: Shot gather and f-k spectrum before and after applying a high-pass velocity filter of 3000 m/s

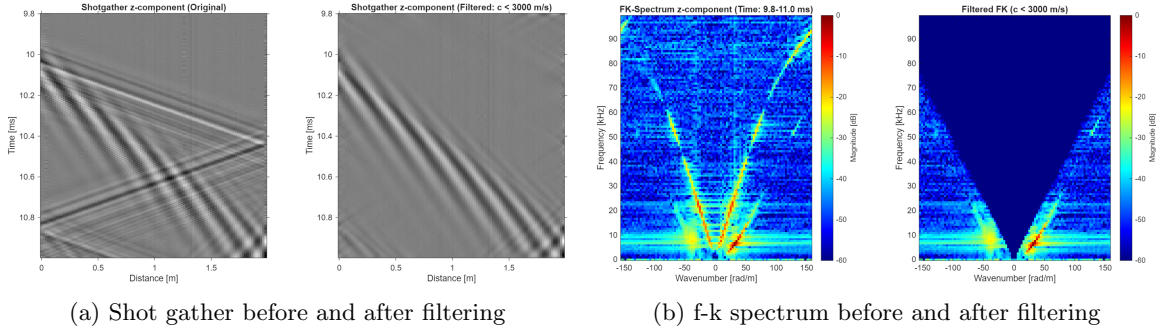


Figure 11: Shot gather and f-k spectrum before and after applying a low-pass velocity filter of 3000 m/s

3.2.4 FC-Spectra

I think I will skip this part and don't put the dispersion curves in here. (Q: tell me if you would like to see them, but the report is getting a bit long with so many plots).

3.2.5 Velocity Analysis

To determine the velocity of the longitudinal and flexural waves, we fitted the highest amplitude of the x-component (see Fig. 12a) and the first arrival of the y-component (see Fig. 12a) on the first and last traces. This determined the velocities corresponding to the red and green slopes, respectively. Using the formula $c = (x_1 - x_0)/(t_1 - t_0)$, where x_0 and x_1 are the locations and t_0 and t_1 are the times of the pick at the first and last points, respectively, we calculated the velocity of the longitudinal wave to be

$$v_l \approx (1.98 \text{ m} - 0 \text{ m}) / (10.45 \text{ ms} - 10.05 \text{ ms}) = 1.98 \text{ m} / 0.40 \text{ ms} = 4950 \text{ m/s}.$$

The calculated velocity for the flexural wave is

$$v_f \approx (1.98 \text{ m} - 0 \text{ m}) / (10.68 \text{ ms} - 10.00 \text{ ms}) = 1.98 \text{ m} / 0.68 \text{ ms} = 2910 \text{ m/s}.$$

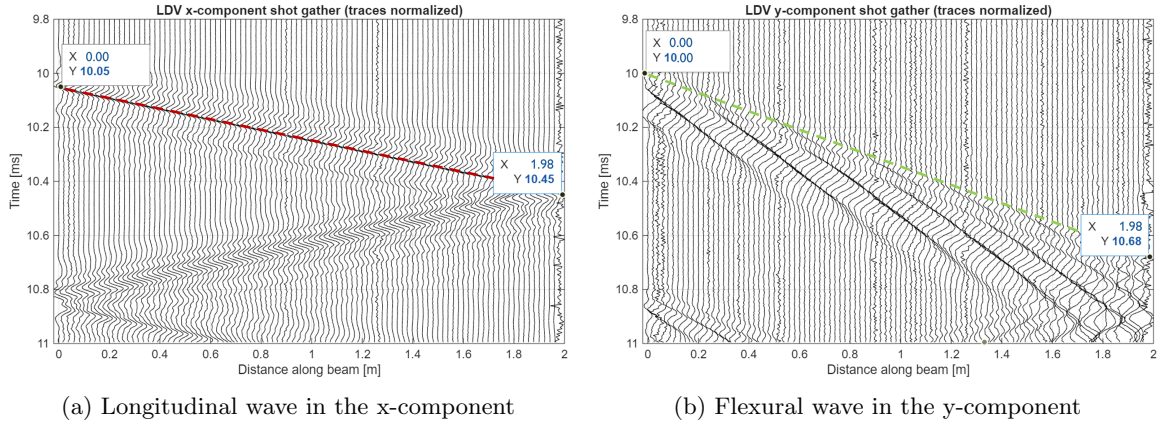


Figure 12: Velocity estimations for the longitudinal wave (red) on the x-component shot gather and the flexural wave (green) on the y-component shot gather. The points picked for determining the velocity are marked.

4 Discussion and Conclusion

Maybe split the parts into just discussion and conclusion like in a paper. Discussion: Why did the reflection cause a surface wave only on one side? What other analyses steps could be done: Make a polarization analysis and look closer at the phases and how they are shifted. Further analysis of the phase spectra are unwrapping the phase. Another way of interesting visualization could be to depict each trace and below the amplitude and phase spectra of each trace like shown by (Features of phase spectrum and its calculation in seismic data processing Mitrofanov... 2012) [Mitrofanov and Priimenko \(2012\)](#).

5 Outlook

Discussion and Outlook to Master thesis project

During the master thesis a variety of fibers will be tested. The experiment will be conducted on different materials like metal beams and concrete as well as on loose, granular material like sand or gravel. By comparing the DAS strain rate with the ground velocities by the LDV transfer functions will be derived for each case and compared. Multiple

References

- Olav Cornelius. Ldv-signal-analysis, 2026. URL <https://github.com/olav-cor/LDV-Signal-Analysis.git>. Accessed: 13.02.2026.
- Robert Ernst, F Zwimpfer, and Jürg Dual. One sensor acoustic emission localization in plates. *Ultrasonics*, 64:139–150, 2016.
- Georgy Mikhailovich Mitrofanov and Viatcheslav Ivanovich Priimenko. Features of phase spectrum and its calculation in the seismic data processing. *Brazilian Journal of Geophysics*, 30(1):15–29, 2012.
- Wenbo Wu, Zhongwen Zhan, Mark Panning, and Andrew Klesh. Fiber seismic network on the moon. *Seismological Research Letters*, 95(4):2153–2163, 2024.

Data Management Plan

Data management: Short (max. 1 page) data management description

Appendix

A LDV Plots

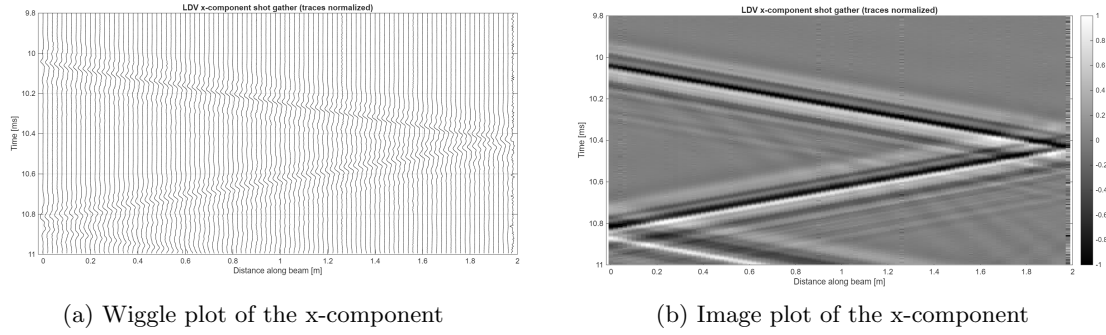


Figure 13

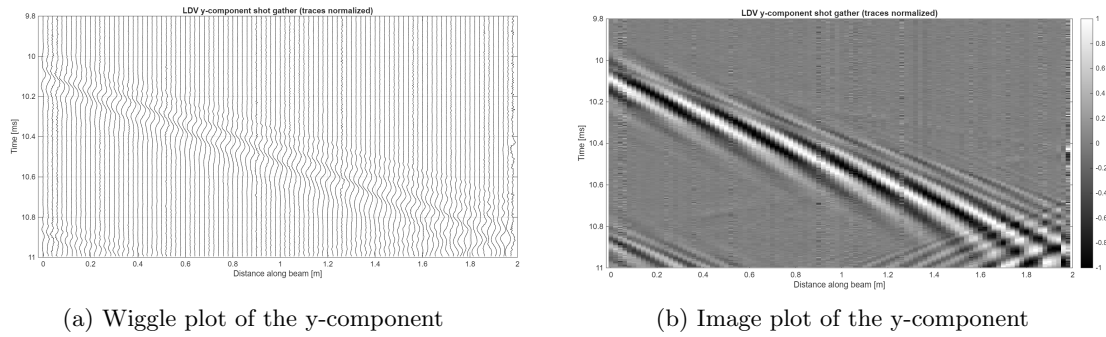


Figure 14

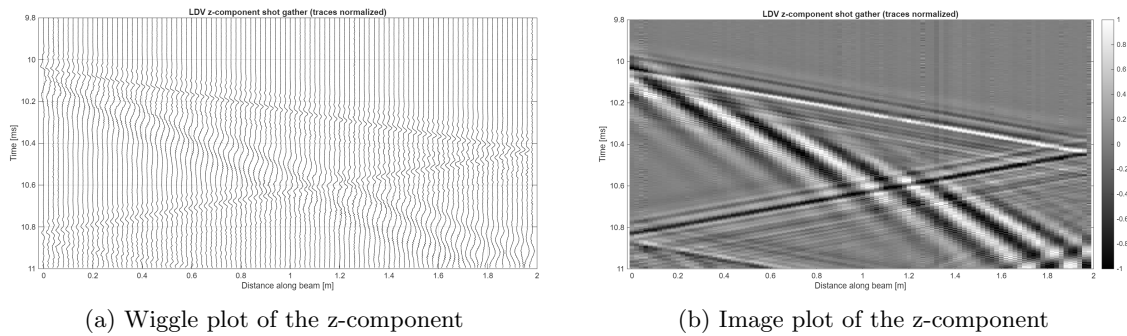


Figure 15

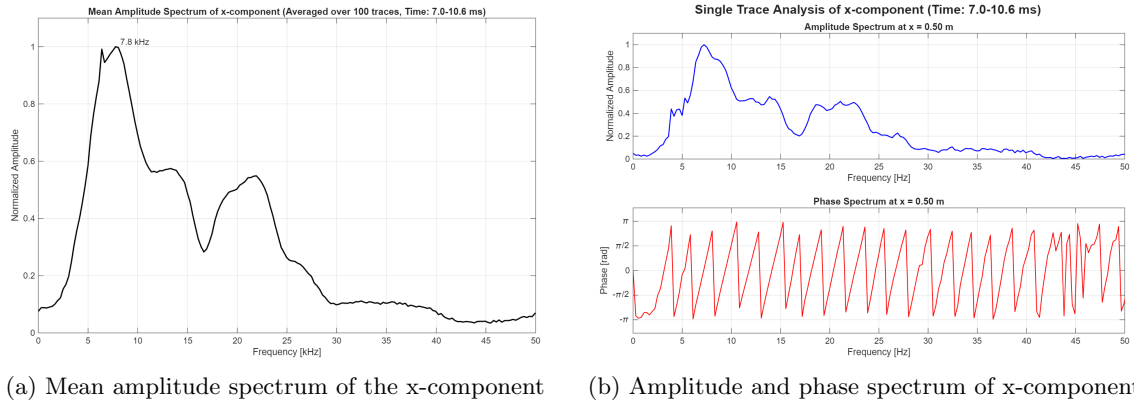


Figure 16

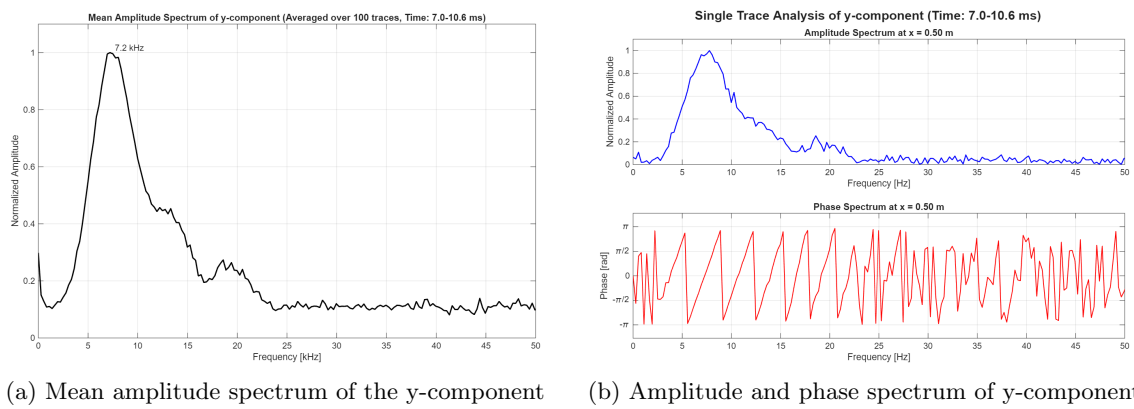


Figure 17

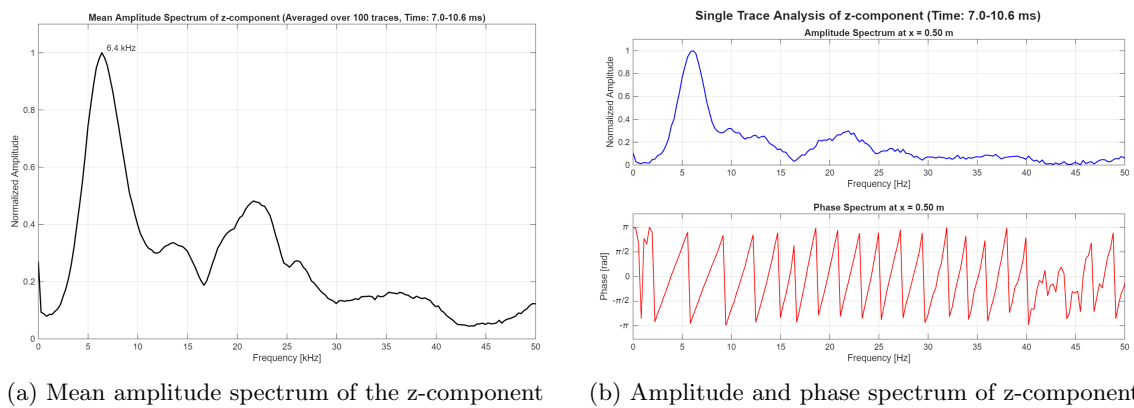


Figure 18

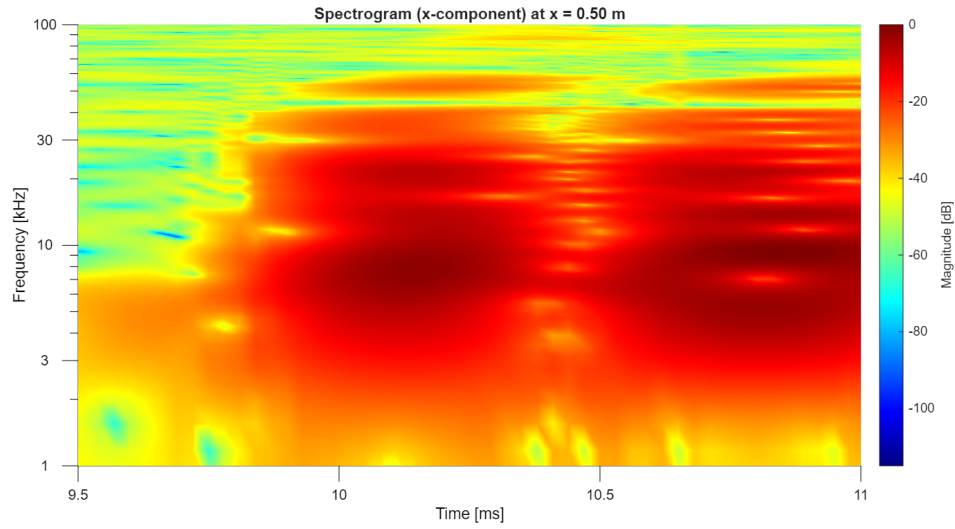


Figure 19: Spectrogram of the x-component at 0.5 m from 9.5 to 11 ms

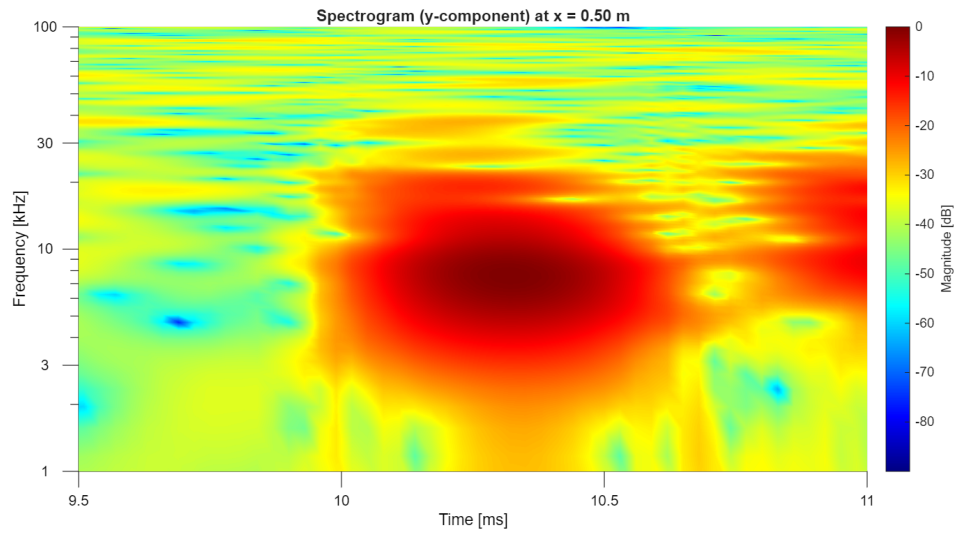


Figure 20: Spectrogram of the y-component at 0.5 m from 9.5 to 11 ms

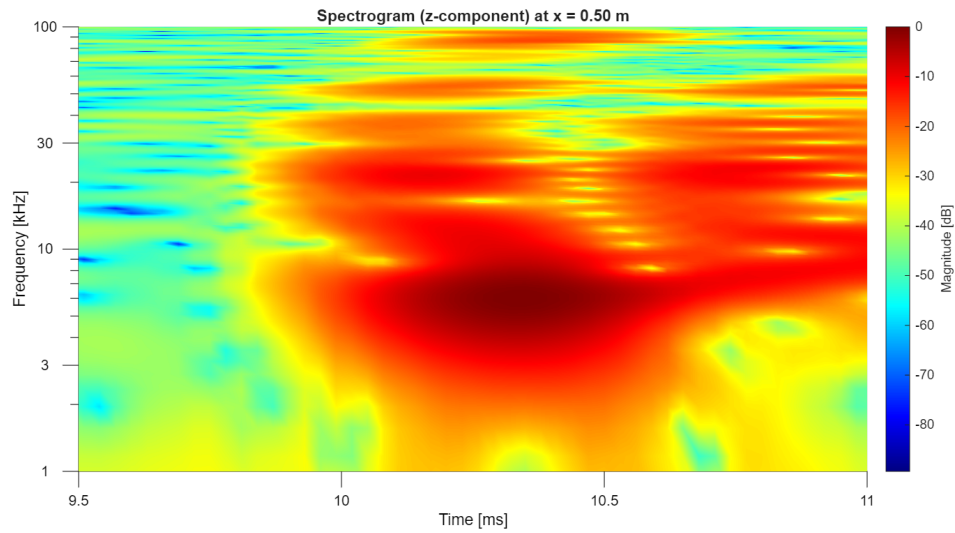


Figure 21: Spectrogram of the z-component at 0.5 m from 9.5 to 11 ms

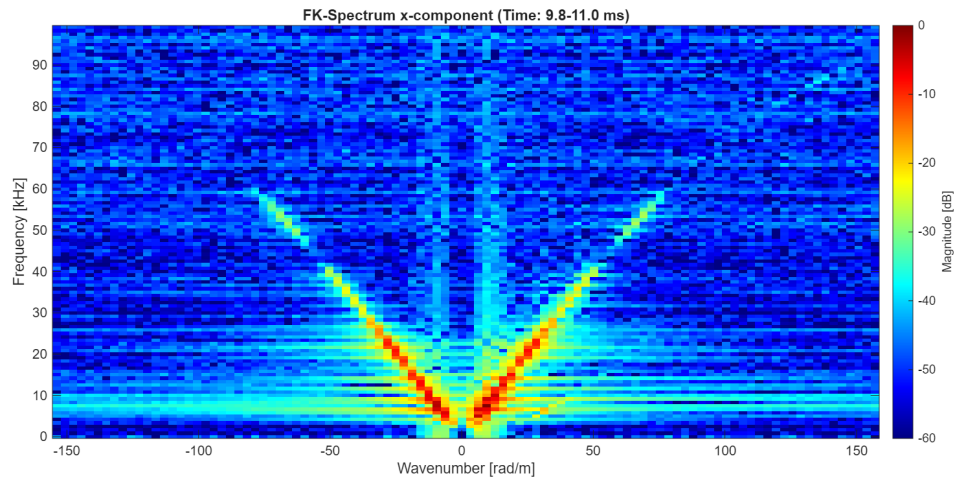


Figure 22: f-k spectrum of the x-component from 9.8 to 11 ms

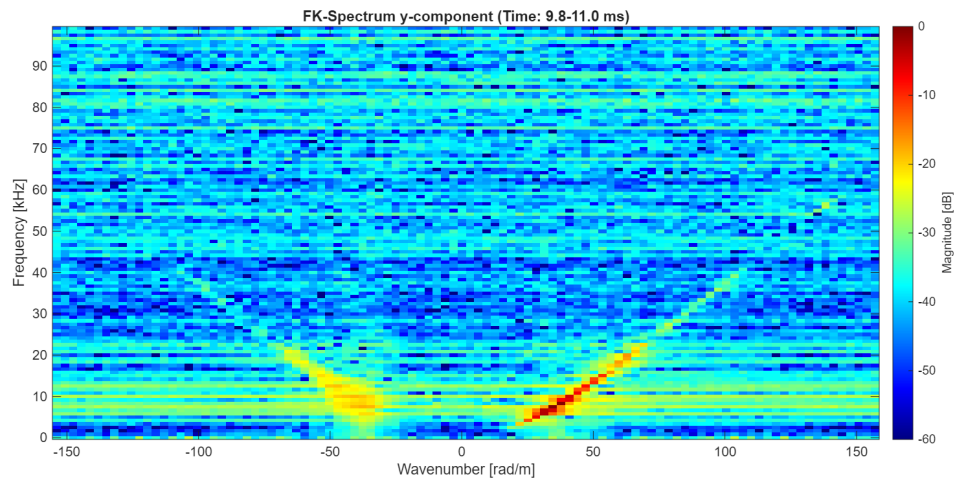


Figure 23: f-k spectrum of the y-component from 9.8 to 11 ms

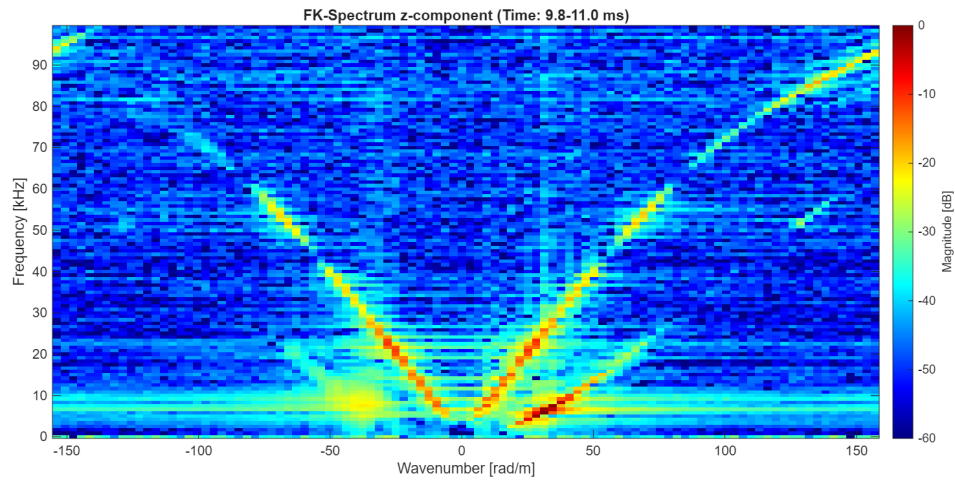


Figure 24: f-k spectrum of the z-component from 9.8 to 11 ms